

Presentation script: "Dataset Generation" (IABR-v2)

How to use this file. Each slide below has: the image to show (in `slides/`), what to point at, and the words to say in simple English. Words in *italics* are stage directions. Total time: about 12 to 15 minutes for 16 image slides + 3 text slides.

One-line summary you can open with:

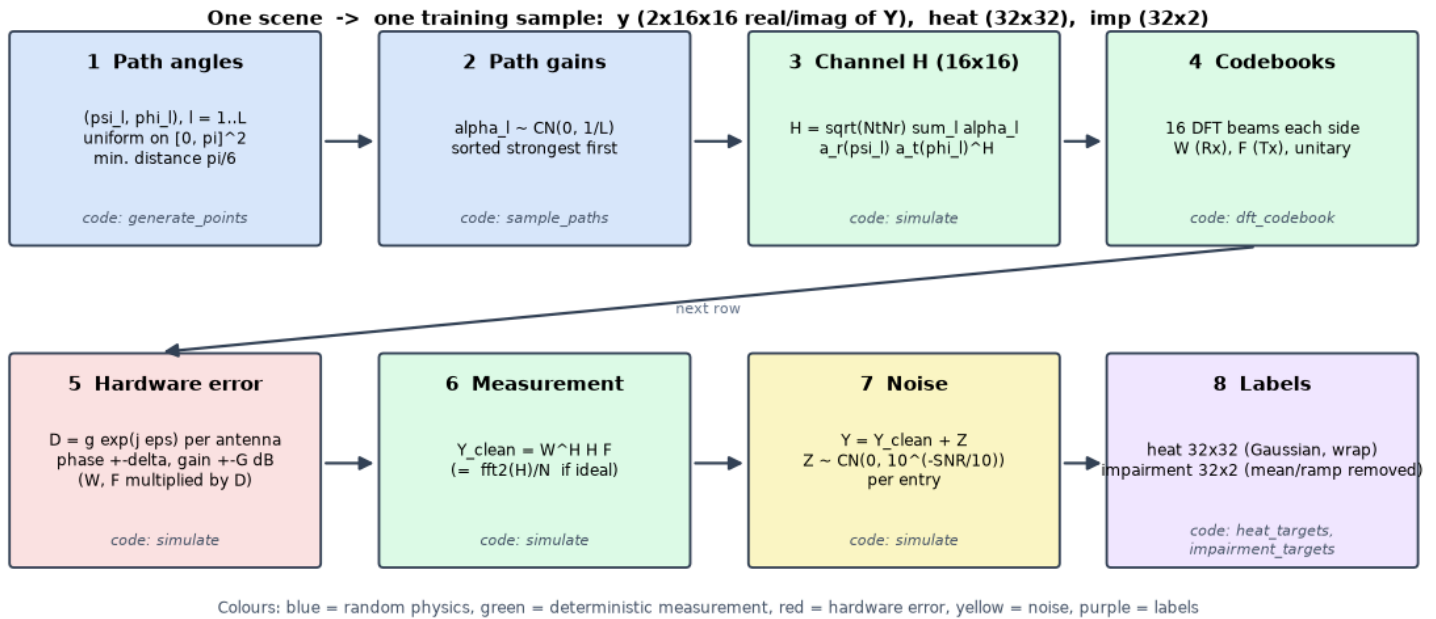
"We do not have a fixed training dataset. We have a *simulator* that makes a new random scene every time. Each scene gives the network one input, the measurement Y , and two answers, a heatmap of path directions and a vector of hardware errors."

Slide 0 - Title / big picture (no image, 30 seconds)

Show: title "Dataset generation for IABR-Net v2".

Say: - "The task is joint AoA and AoD estimation for a millimetre-wave MIMO link. Both sides have a 16-antenna array." - "The receiver sends known probing beams and measures a 16 by 16 matrix Y . From Y , the network must find the arrival angle (AoA) and departure angle (AoD) of every path." - "Our data is not downloaded. It is generated by a physical simulator. Training data is an endless random stream. Test data is a set of frozen files." - "I will now walk through the generation from the first random number to the final label."

Slide 1 - The 8-step flow (S01_flow_8_steps.png)

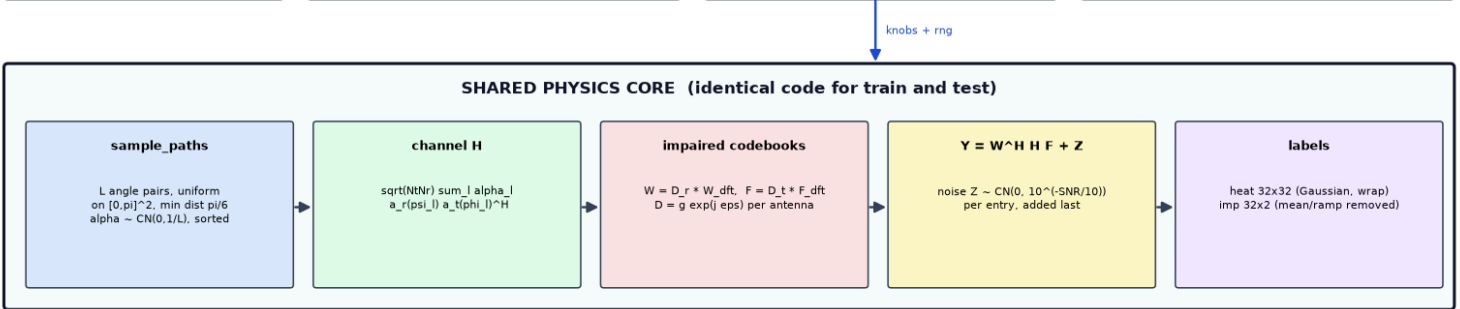
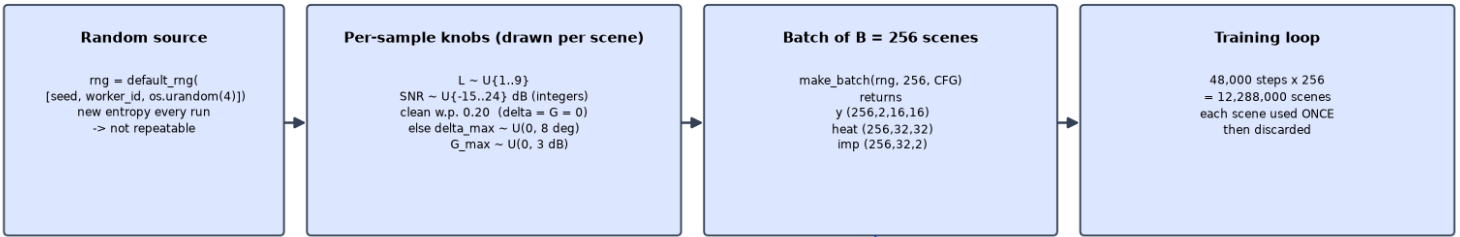


Point at: the 8 boxes, left to right, top row then bottom first. Colours: blue = random physics, green = fixed maths, red = hardware error, yellow = noise, purple = labels.

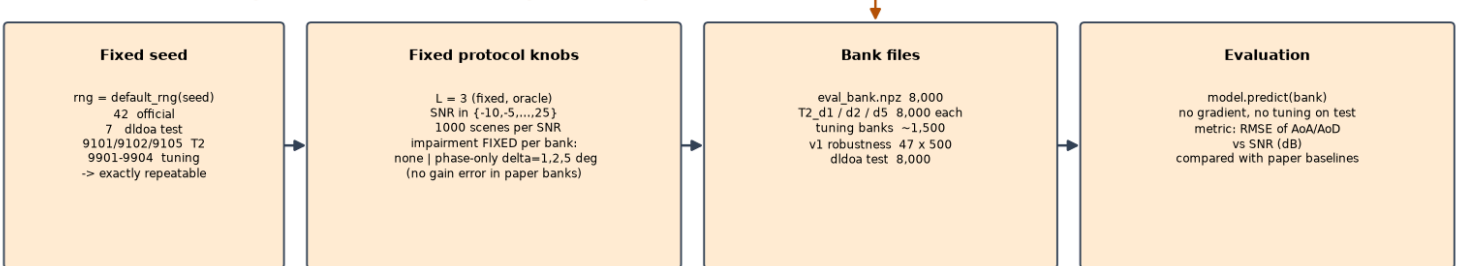
Say: - "This is the whole pipeline. One scene goes through eight steps." - "Step 1 and 2 are random: we pick the path directions and the path strengths." - "Step 3 builds the channel matrix H from those paths." - "Step 4 is the set of beams used for measuring, the DFT codebooks. Step 5 adds hardware errors to those beams. Step 6 measures: Y equals W -Hermitian times H times F ." - "Step 7 adds noise at the chosen SNR." - "Step 8 makes the answers: a 32 by 32 heatmap and an impairment vector." - "The network only ever sees the result of step 7. Steps 1 to 5 are hidden from it. Step 8 is what it must predict."

Slide 2 - Training vs testing pipeline (S02_train_vs_test_pipeline.png)

TRAINING (endless fresh stream, nothing stored)



TESTING (frozen banks, generated once, saved to .npz, read-only)

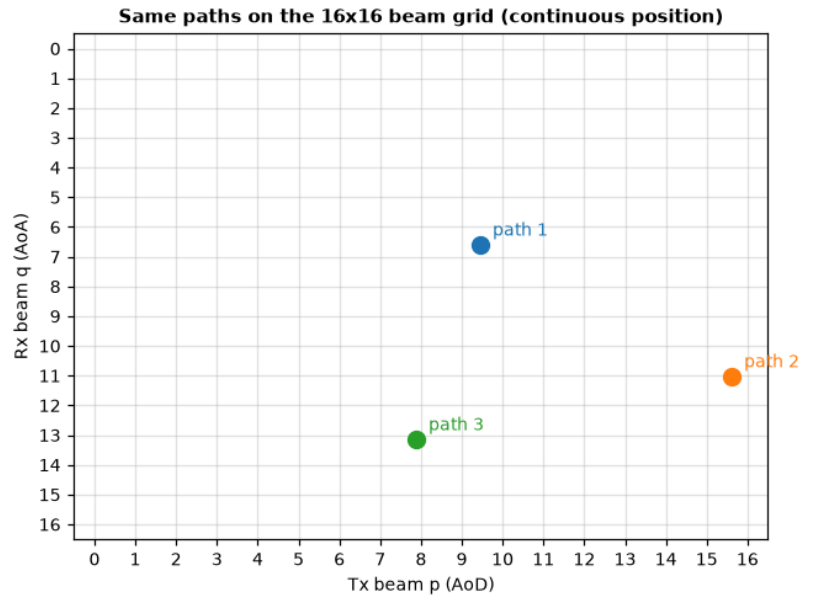
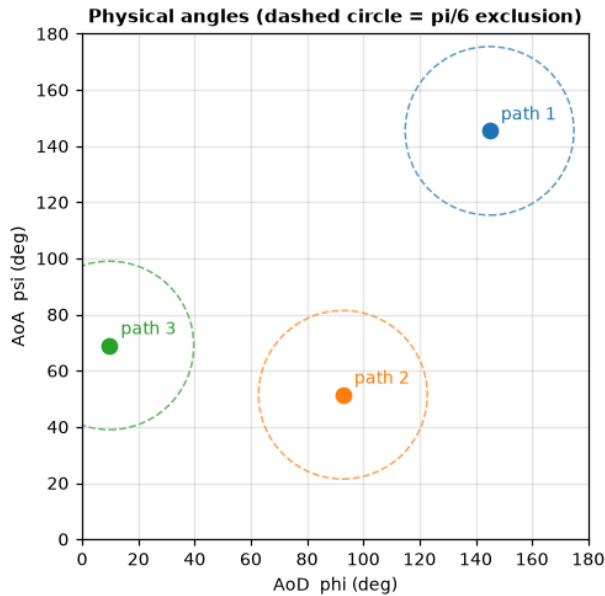


No scene is shared between training (random, endless) and test (frozen seeds): leakage-free by construction.

Point at: the blue top lane, the grey middle box, the orange bottom lane.

Say: - "The middle box, the physics core, is the same code for training and testing. Only the *knobs* and the *random seeds* are different." - **Training (top):** - "Every batch is created on the fly. The random generator is seeded with the seed, the worker id and 4 random bytes from the operating system, so each run is different." - "For each scene we draw: number of paths L from 1 to 9, SNR from minus 15 to 24 dB (whole numbers), and with probability 20 percent the hardware is perfect. Otherwise the phase-error limit is drawn from 0 to 8 degrees and the gain-error limit from 0 to 3 dB." - "A batch is 256 scenes. Training runs 48,000 steps. That is 12.29 million scenes. Each scene is used once and thrown away. Nothing is stored." - **Testing (bottom):** - "Test sets are generated once with a fixed seed and saved to files. So anyone can reproduce them exactly." - "In the test sets L is always 3, SNR goes from minus 10 to 25 in steps of 5 dB, with 1,000 scenes per SNR, so 8,000 scenes. The hardware error is fixed for each bank: none, or phase-only with 1, 2 or 5 degrees." - "Training scenes and test scenes never overlap. Test seeds are fixed (42, 7, 9101, 9102, 9105). Training seeds come from the operating system."

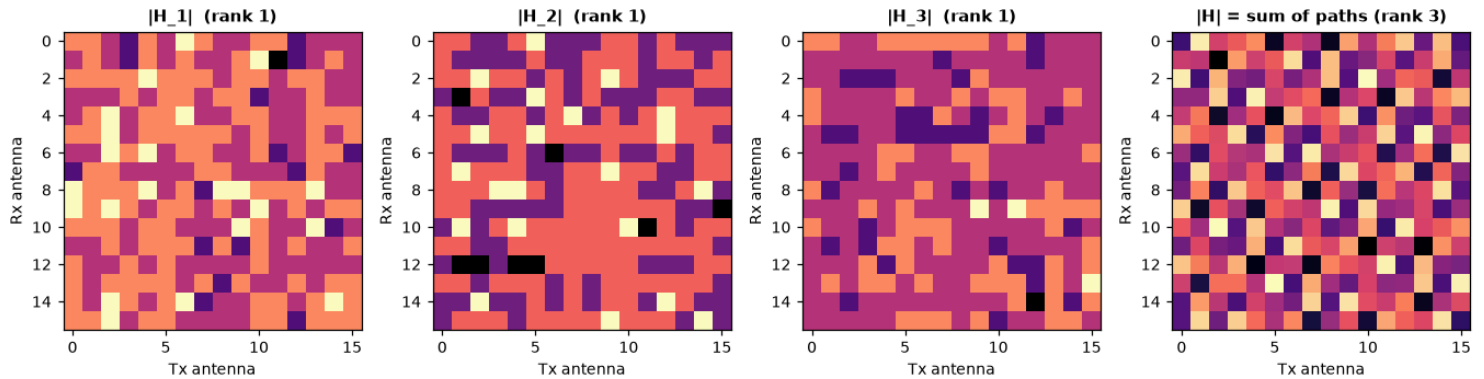
Slide 3 - Step 1: path angles (S03_path_angles.png)



Point at (left): three dots and dashed circles. **Point at (right):** the same three paths on the 16 by 16 beam grid.

Say: - "A path is one way the signal travels from transmitter to receiver. Each path has a departure angle ϕ (AoD) and an arrival angle ψ (AoA)." - "We draw the pairs uniformly on 0 to 180 degrees on both axes. This is the authors' rejection sampler: a new path is rejected if it is closer than $\pi/6$ (30 degrees) to an earlier one, measured as the ordinary distance in the (ϕ , ψ) plane. The dashed circles show that exclusion zone." - "The receiver does not see angles directly. It sees them through the beam grid. The position of a path on the grid depends on the *cosine* of the angle, not on the angle." - "The formulas: row q equals 16 times (minus $\pi \cos \psi$, wrapped into 0 to 2 π) divided by 2 π . Column p equals 16 times ($\pi \cos \phi$, wrapped) divided by 2 π . Rows are AoA, columns are AoD." - "Important side effect: because of the cosine, paths near 0 or 180 degrees, called end-fire, get squeezed together. Later I will show numbers for that."

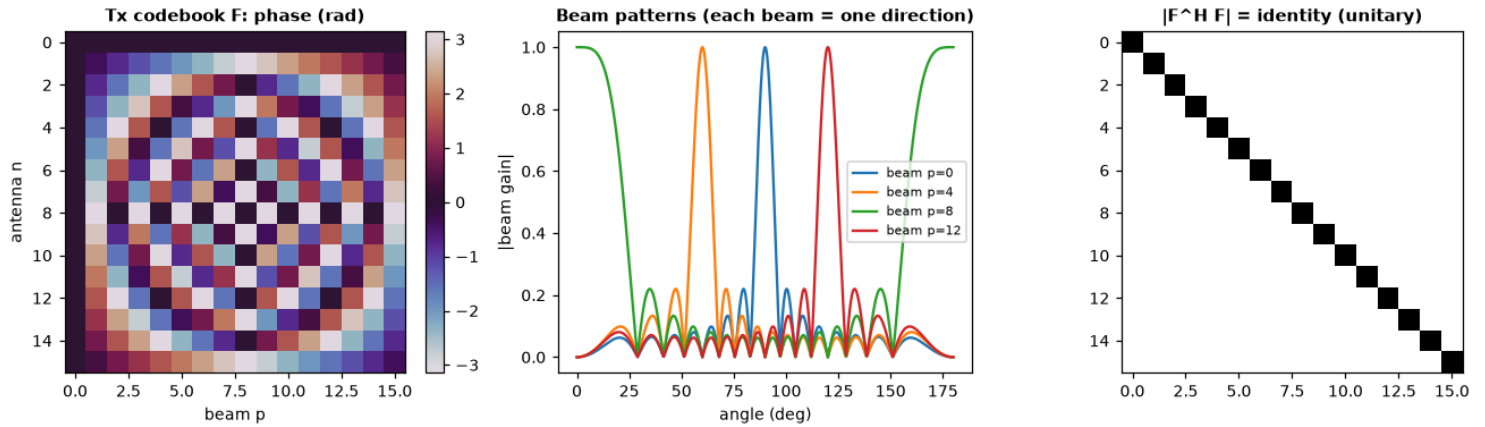
Slide 4 - Steps 2 and 3: path gains and the channel H (S04_channel1_H.png)



Point at: the three left panels (one path each, rank 1) and the right panel (sum, rank 3).

Say: - "Step 2: every path gets a complex gain α , drawn from a complex Gaussian with variance 1 over L . So the total average power is 1. We sort the gains, strongest first." - "Step 3: the channel matrix H is the sum over paths: square root of (N_t times N_r), times α , times the receive steering vector, times the conjugate transpose of the transmit steering vector. Here N_t and N_r are both 16, so the factor is 16." - "The steering vector for angle θ is $\exp(-j \pi k \cos \theta)$ divided by root 16, for antenna k . It is a half-wavelength uniform line array." - "Each single path is a rank-one matrix. Its magnitude is flat across the array. Only the phase slope carries the angle. The sum of three paths has rank 3. You can see three non-zero singular values in the notebook."

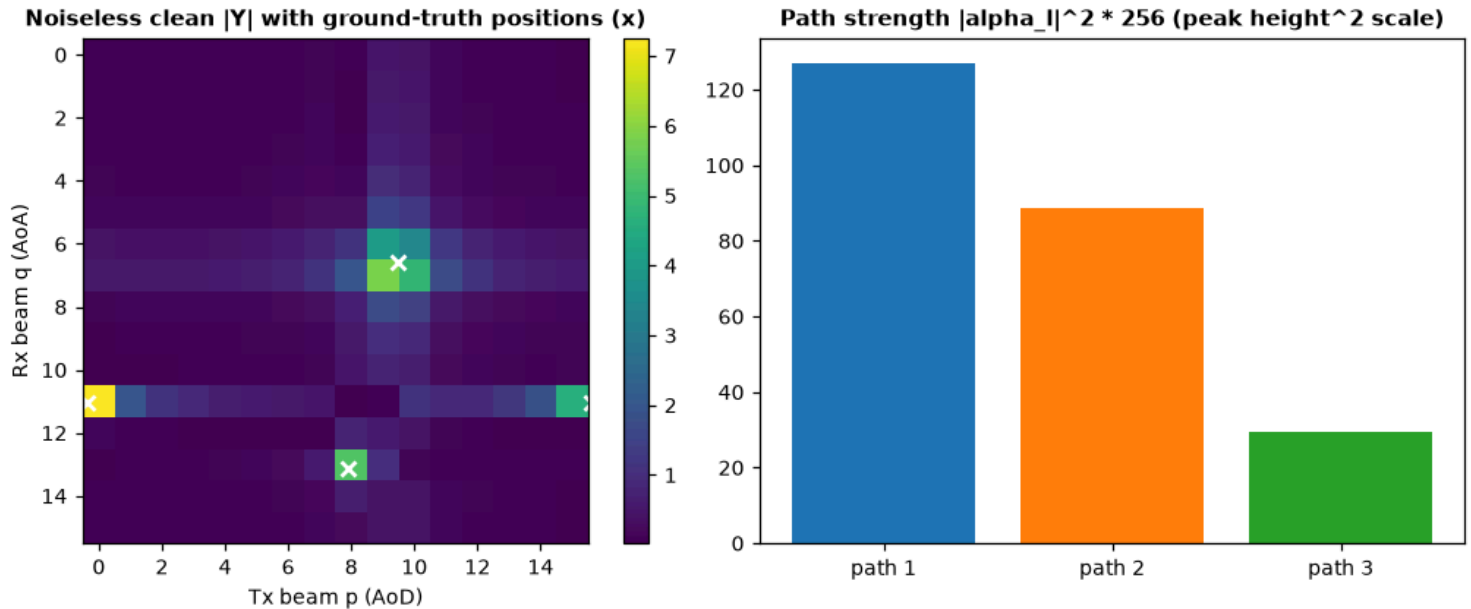
Slide 5 - Step 4: the DFT codebooks (S05_codebooks.png)



Point at: left = phase pattern of the transmit codebook, middle = beam patterns, right = identity matrix.

Say: - "The receiver does not read H antenna by antenna. It uses 16 fixed beams on each side. These are DFT beams, equally spaced in cosine of angle." - "F is the transmit codebook, W is the receive codebook. Each is a 16 by 16 matrix. Each column is one beam." - "The middle plot shows that each beam points in one direction." - "The right plot shows $F^H H F$ is the identity. So the codebooks are unitary: no gain, no loss, no correlation between beams. This matters because it makes the noise white and the SNR easy to define."

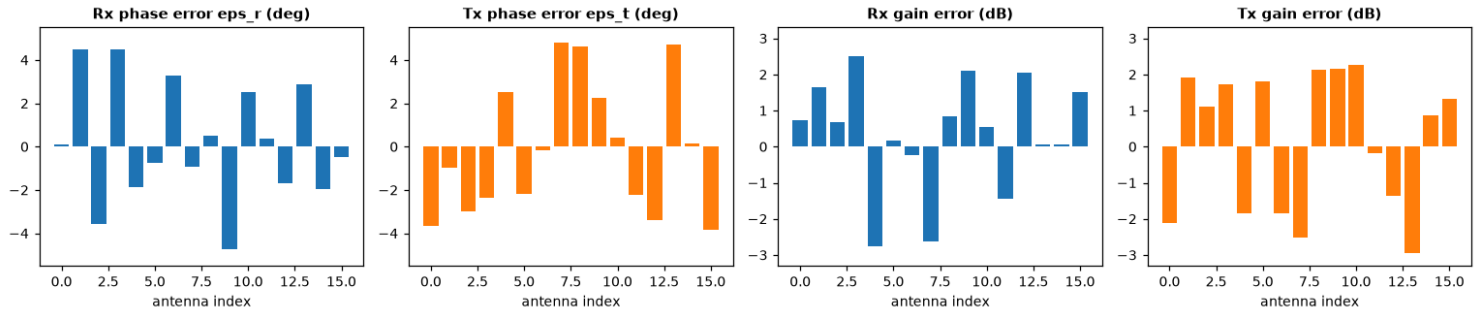
Slide 6 - Step 6 (ideal): the clean measurement Y (S06_clean_measurement_Y.png)



Point at (left): the bright spots and the white crosses (ground truth). **Point at (right):** bar heights = path strength.

Say: - "The measurement is Y equals W -Hermitian times H times F . Y is a 16 by 16 complex matrix. Entry (q, p) is the response of receive beam q and transmit beam p ." - "With ideal hardware this is *exactly* a two-dimensional FFT of H divided by 16. We checked: the difference is about 10 to the power minus 14." - "So every path shows up as a peak at its beam position. The white crosses are the true positions. A path that falls between two beams leaks into its neighbours, that is why the peak looks like a small cross." - "Strong paths give tall peaks; weak paths give small ones. That is the right-hand chart."

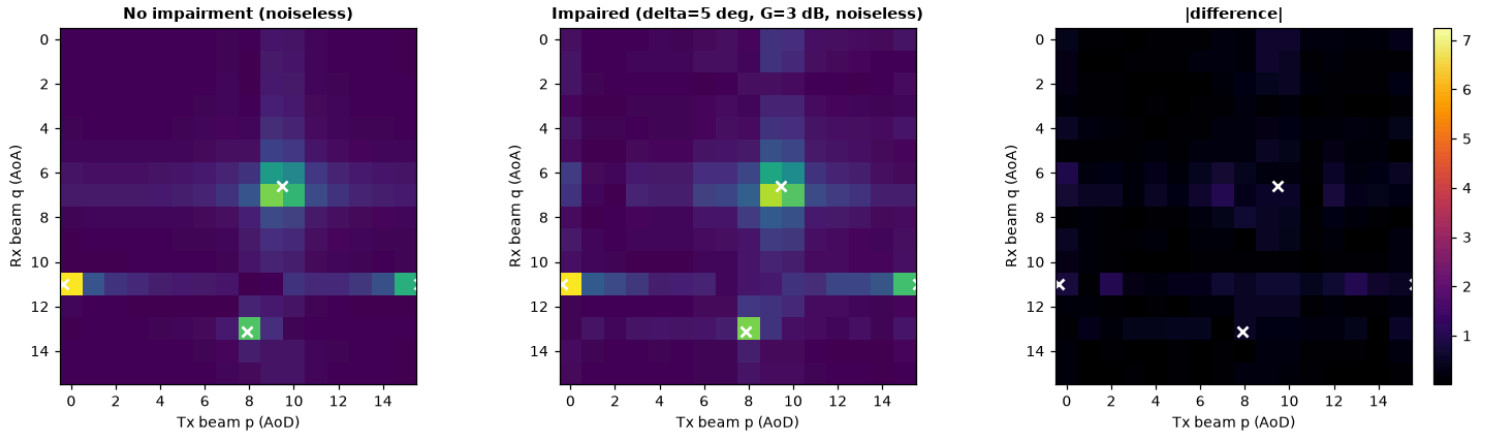
Slide 7 - Step 5: hardware errors per antenna (S07_per_antenna_errors.png)



Point at: the four bar charts: receive phase, transmit phase, receive gain, transmit gain.

Say: - "Real arrays are not perfect. Every antenna has a small phase error and a small gain error." - "We model each antenna weight as multiplied by D equals g times e to the j epsilon. This is applied to both codebooks: W becomes D -receive times W , F becomes D -transmit times F ." - "The phase error epsilon is uniform between minus delta and plus delta degrees. The gain error is uniform in decibels between minus G and plus G ." - "In this demo scene the limits are the upper end of the training range (up to 8 degrees, 3 dB). Each of the 32 antennas (16 receive plus 16 transmit) has its own random error. The same error is used for all beams. A new error is drawn for every scene." - *Be ready to say:* "The phase part follows the base paper. The gain part is our own extension. The paper models phase only."

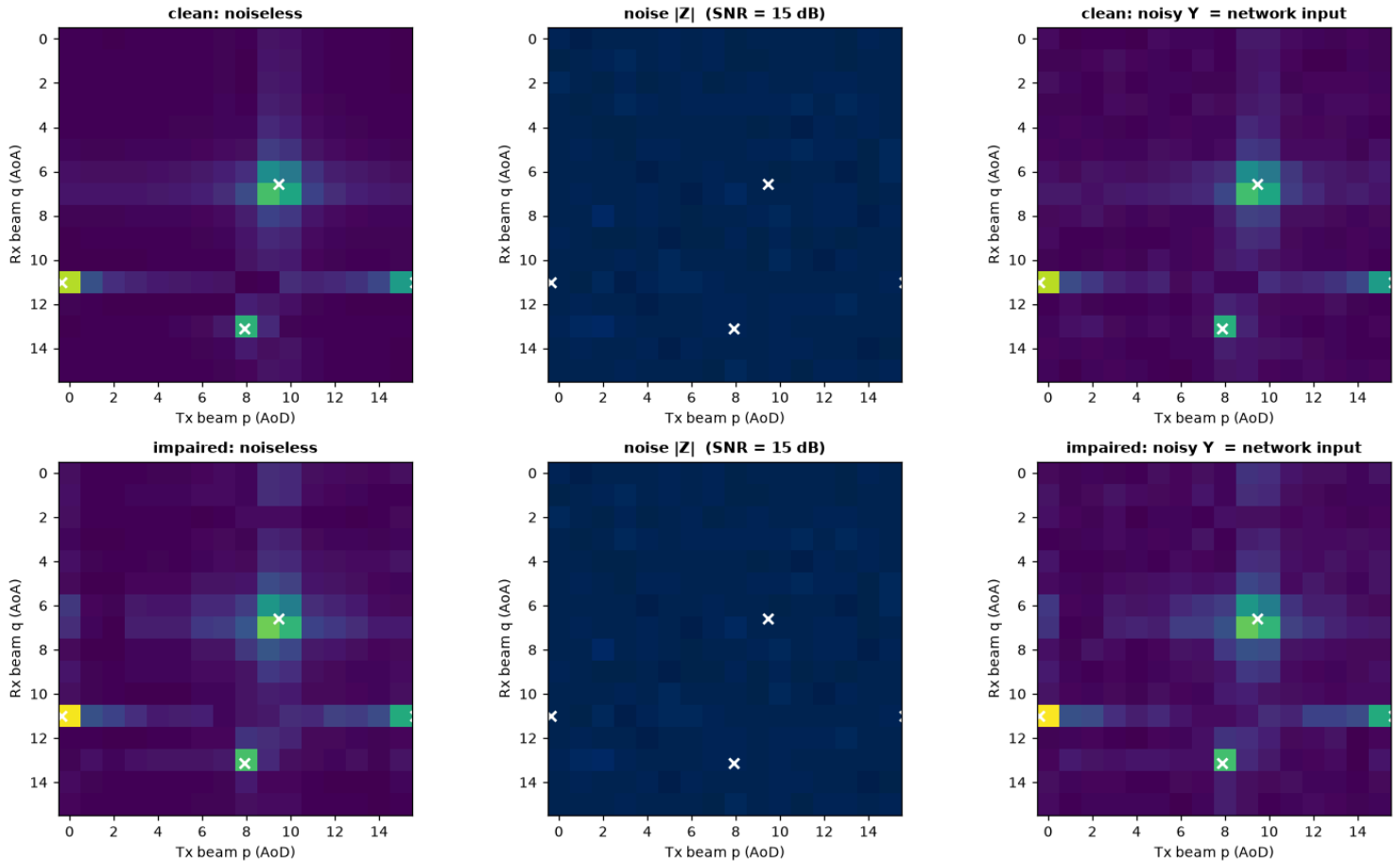
Slide 8 - Clean vs impaired measurement (S08_clean_vs_impaired.png)



Point at: left = no impairment, middle = impaired, right = difference.

Say: - "Same scene, same paths. Left: perfect hardware. Middle: with hardware errors switched on." - "The peaks stay in the same place. But energy leaks along the row and column of each peak, and the peak heights change." - "The right panel is the difference. In this example it is about 32 percent of the signal norm (ratio 0.316, or minus 10 dB)." - "Small errors, like 1 or 2 degrees of phase, change Y much less, so the hard cases are the larger mixed ones. That is why the network has to learn to undo the impairment."

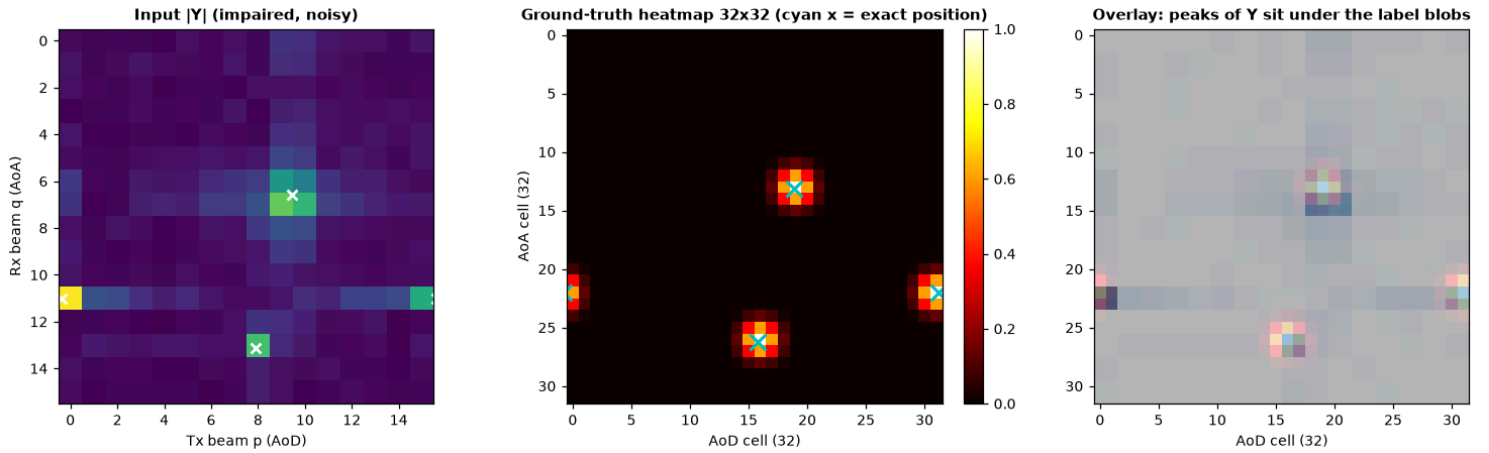
Slide 9 - Step 7: adding noise (S09_noise_2x3.png)



Point at: top row = clean hardware, bottom row = impaired hardware. Columns: noiseless, noise, noisy.

Say: - "Noise is added last, after the receive beams. It is complex Gaussian, independent for every entry of Y. Its power is 10 to the power minus SNR over 10." - "In code: each real and imaginary part has standard deviation square root of (10 to the minus SNR/10, divided by 2)." - "Because the total path power is 1 and the codebooks are unitary, the average signal power per entry of Y is 1. So the nominal SNR is exactly a per-entry ratio. In this example I set 15 dB, and the realised value was 15.3 dB, because this one scene has a slightly different noise draw and its path power sums to 0.958 instead of 1." - "Top and bottom rows show the same scene with and without hardware error, so you can compare the effect of the error directly." - "The right column is the final network input."

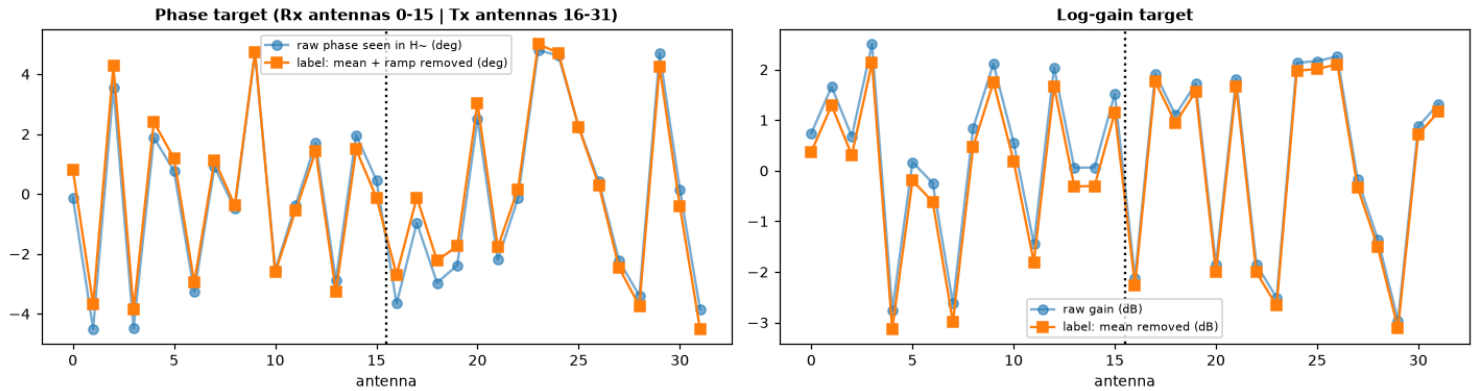
Slide 10 - Step 8a: heatmap ground truth (S10_heatmap_ground_truth.png)



Point at: left = input Y, middle = 32x32 heatmap, right = overlay.

Say: - "First answer the network must produce: a 32 by 32 heatmap. It is twice finer than the 16 by 16 measurement, so the network can place a path *between* two beams." - "For every path we compute its exact position on the 32-grid, round to the nearest cell, and put a Gaussian blob there. Width is one cell. The peak value is exactly 1." - "The distance is circular, wrap-around, because beam space repeats. In this example one path sits at the edge and its blob continues on the opposite side of the image." - "If several paths overlap, we take the maximum, not the sum. So the heatmap stays between 0 and 1." - "Rows are AoA, columns are AoD. The right panel shows that the peaks of Y sit right under the label blobs."

Slide 11 - Step 8b: impairment ground truth (S11_impairment_ground_truth.png)

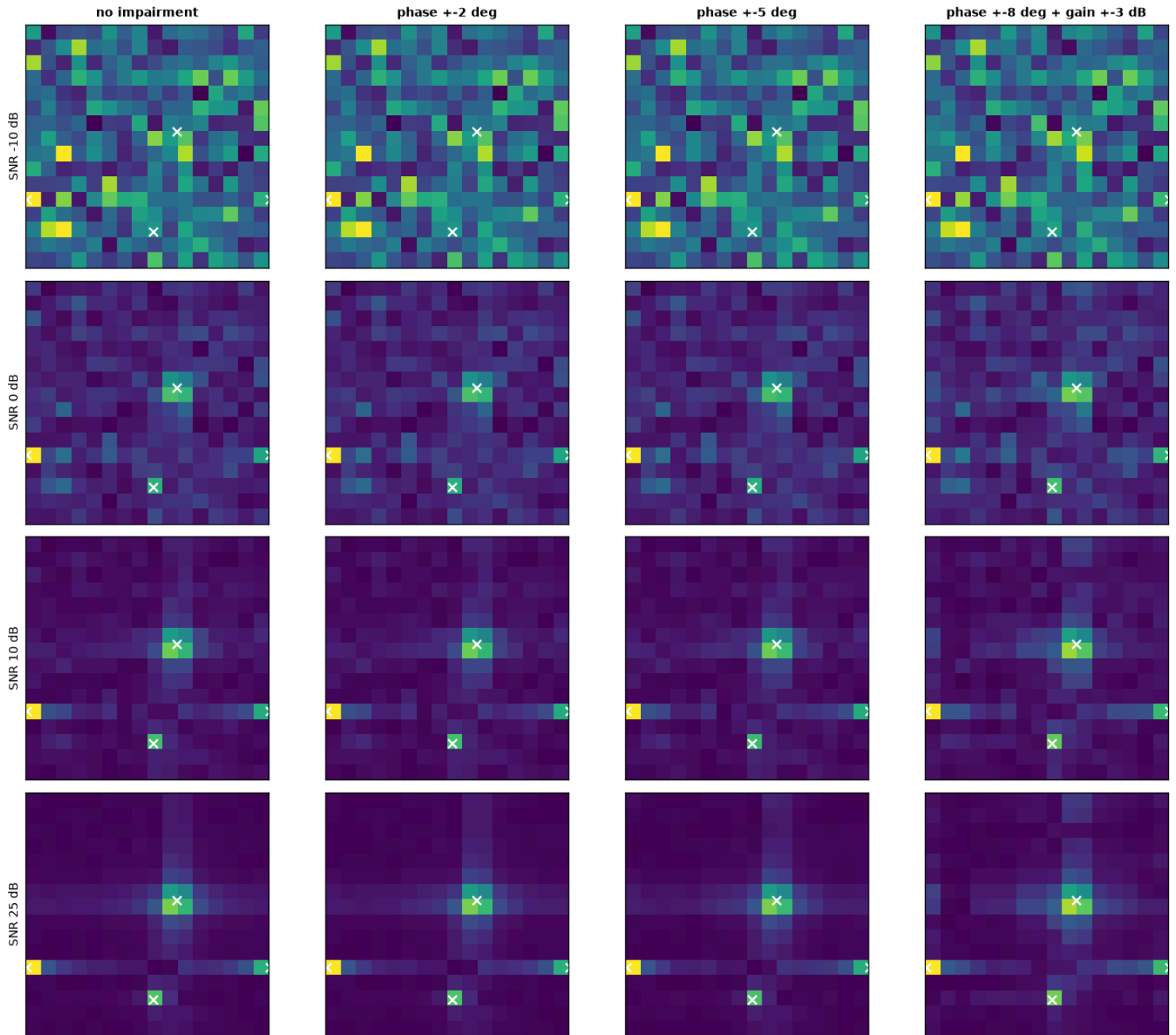


Point at: left = phase (deg), right = log-gain. Dotted line: receive antennas 0-15 on the left, transmit antennas 16-31 on the right. Blue line = raw, orange = label.

Say: - "Second answer: the true hardware error of each of the 32 antennas. It is a 32 by 2 array: phase and log-gain." - "Sign convention: because the measurement acts on H as conjugate of D-receive times H times D-transmit, the receive phase label is minus the receive error angle, and the transmit label is plus the transmit error angle." - "We do *not* ask the network for parts that cannot be identified from data. For phase, we remove the mean and the linear ramp on each side. A linear phase ramp across an array looks exactly like a shift in the arrival angle, so it cannot be told apart from a real angle change. The mean phase is a common offset. For gain we remove the mean." - "So the orange line is the raw blue line minus its mean and slope. We checked in the notebook that the mean and slope of the label are zero."

Slide 12 - Same scene, different SNR and impairment

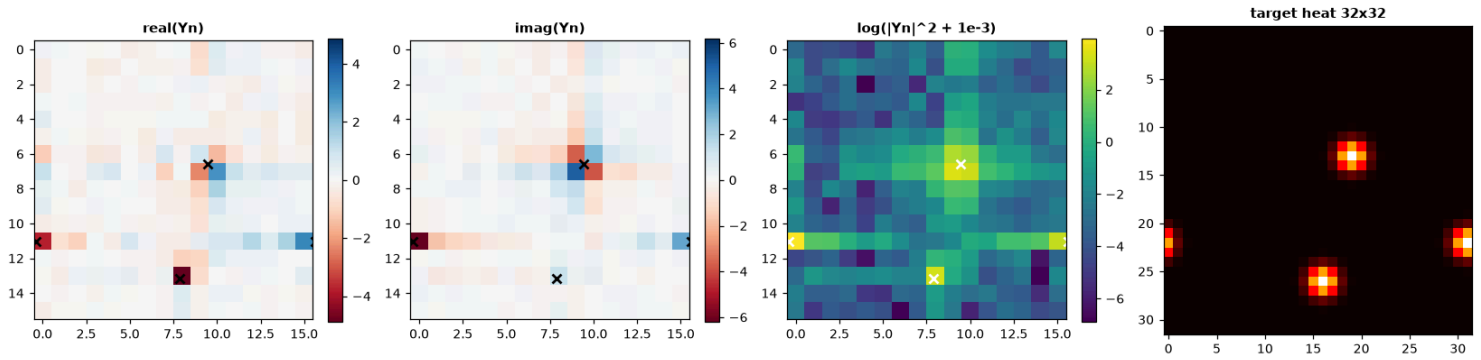
(S12_snr_x_impairment_grid.png)



Point at: rows = SNR (minus 10, 0, 10, 25 dB), columns = none, 2, 5, 8 degrees plus 3 dB gain. White crosses = ground truth.

Say: - "This is the key picture. The same scene is shown at different SNR and different impairment levels." - "Going down (higher SNR), the paths come out of the noise. At minus 10 dB the peaks are almost invisible." - "Going right (more impairment), the leakage around each peak grows. At 25 dB and the last column, the peaks are still in the right places but surrounded by strong leakage." - "The network must cope with both problems at the same time: noise at low SNR and leakage at high SNR."

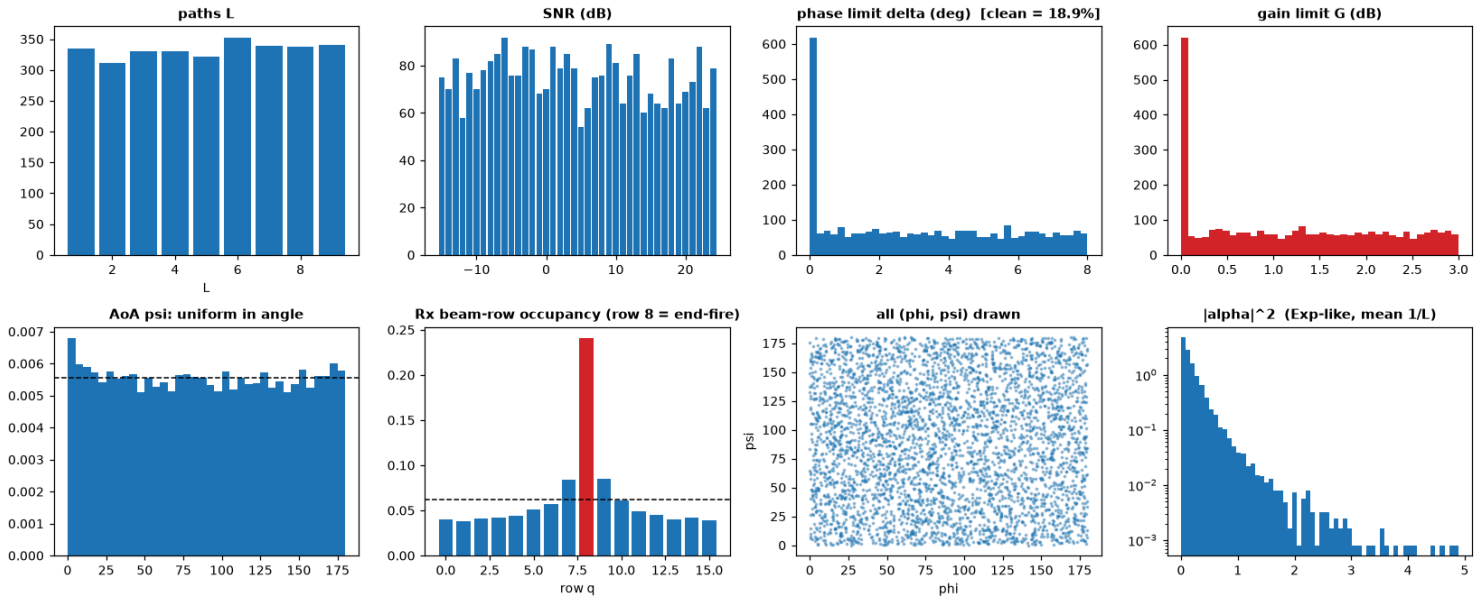
Slide 13 - What the network actually sees (S13_network_input_channels.png)



Point at: four panels: real part, imaginary part, log-magnitude, target heatmap.

Say: - "This part is inside the network, not part of the stored data. The dataset stores Y as two planes: real and imaginary, shape 2 by 16 by 16." - "In the network, Y is first divided by its RMS, so the scale does not depend on SNR. Then a third channel is added: $\log$ of (magnitude squared plus 0.001)." - "The log channel is what makes weak paths visible at low SNR. Look at the third panel: peaks show up strongly there." - "So one training sample is three tensors: **y** of size $2 \times 16 \times 16$, **heat** of size 32×32 , and **imp** of size 32×2 ."

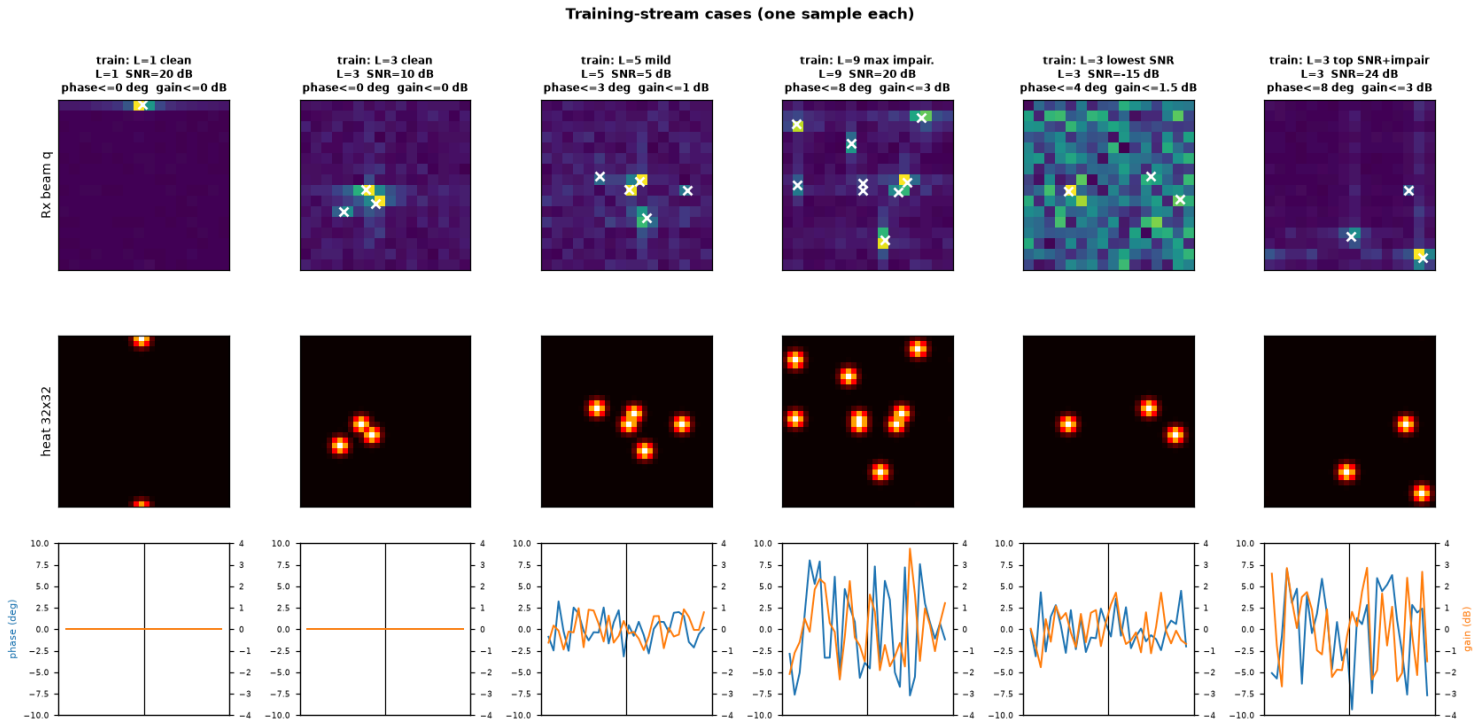
Slide 14 - The training stream as a whole (S14_training_stream_stats.png)



Point at: top row = L, SNR, phase limit, gain limit. Bottom row = AoA histogram, beam-row occupancy, all angle pairs, path power.

Say: - "This is a sample of scenes drawn with the real training recipe." - "Top row: L is uniform from 1 to 9. SNR is uniform from minus 15 to 24. About 20 percent of scenes have zero impairment, that is the tall bar at zero. The rest are uniform up to 8 degrees and 3 dB." - "Bottom row: angles are uniform in angle (left and third plot). But look at the second plot: the *beam row* occupancy is not uniform. The red bar is row 8, the end-fire row. About 22 to 24 percent of all paths fall there, versus 6.25 percent if it were uniform. This is the cosine effect I mentioned: the arcsine law." - "Last plot: path power is exponential-like, as expected from Rayleigh gains. Many weak paths, a few strong ones."

Slide 15 - One sample of every training case (S15_training_cases.png)

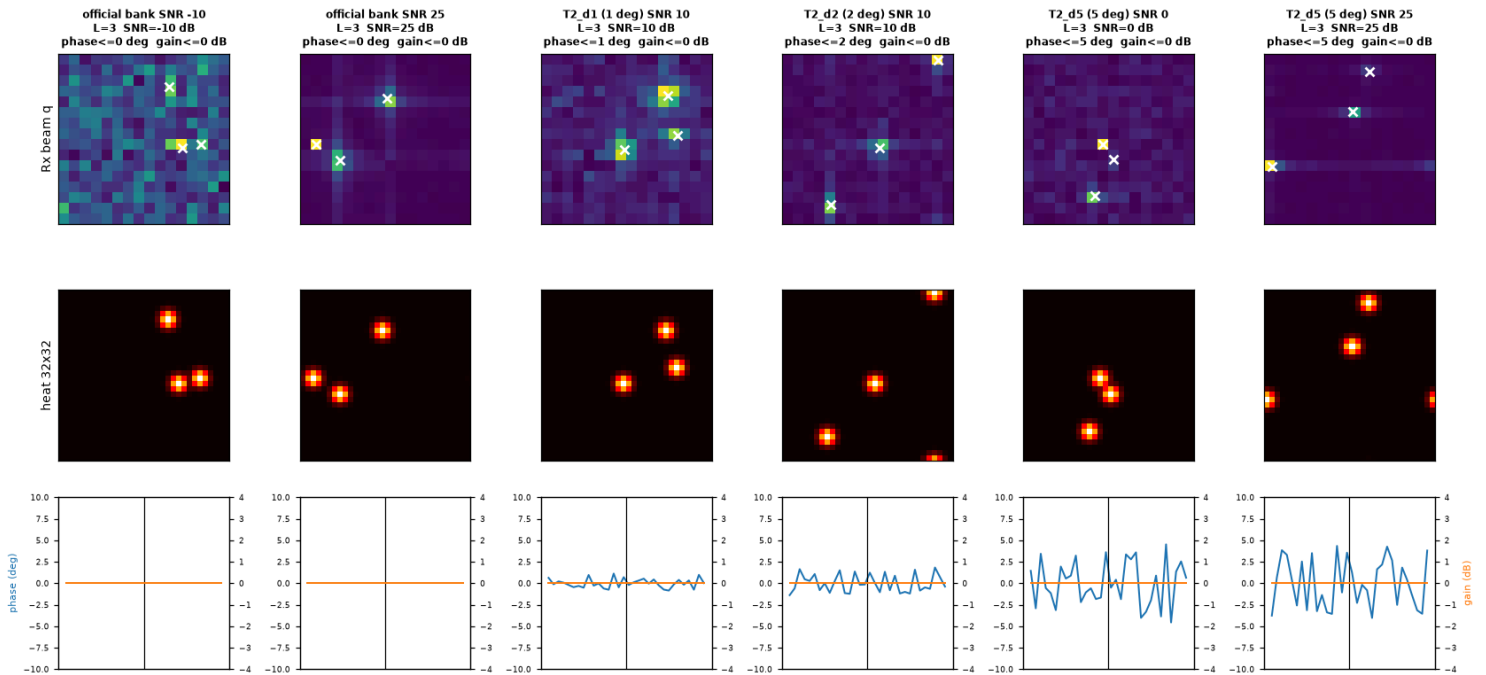


Point at: columns left to right; for each, top = Y, middle = heatmap, bottom = impairment label.

Say: - "One example for each kind of training case: L=1 clean, L=3 clean, L=9 with maximum impairment, L=3 at the lowest SNR (-15 dB), and L=3 at top SNR (24 dB) with impairment." - "With L equals 1 you see a single peak. With L equals 9 there are many overlapping peaks and some blobs in the heatmap come close together." - "At minus 15 dB the input looks almost like noise, but the label is still clean blobs. That gap is what the network learns to close." - "The bottom row shows the impairment label. Flat for the clean cases, larger and noisier for the impaired ones."

Slide 16 - One sample of every test case (S16_test_cases.png)

Test-bank cases (L = 3, paper protocol; one sample each)



Point at: official bank at SNR -10 and 25; T2 banks with 1, 2, 5 degrees.

Say: - "These are the test-bank settings, one example each: official bank at SNR -10 and 25; T2 banks with 1, 2 and 5 degrees of phase error. L is always 3. The official bank has no impairment. The T2 banks have phase-only impairment." - "At 1 and 2 degrees the picture is close to clean. The 5 degree bank is where the impairment matters most." - "In the test banks the gain error is zero, exactly like the paper. The gain error is only used in our own stress tests."

Slide 17 - Test banks table (no image, or put this table on a slide)

Bank	Size	L	SNR (dB)	Hardware error	Seed
official (eval_bank.npz)	8,000	3	-10 to 25, step 5 (1,000 each)	none	42
dl doa test	8,000	3	same	none	7
T2_d1 / d2 / d5	8,000 each	3	same	phase only, 1 / 2 / 5 deg	9101 / 9102 / 9105
tuning banks	about 1,500 total	mixed	mixed	mixed	9901 to 9904
v1 robustness	47 banks x 500	3	various	phase / gain / SNR sweeps	various
training	12,288,000 fresh scenes	1 to 9	-15 to 24	20% none, else up to 8 deg and 3 dB	OS entropy

Say: - "The official bank is reproduced bit-exactly from the authors' generator with seed 42. A second independent draw with seed 7 is also bit-exact." - "Tuning banks are separate from the final test banks. We tune the self-calibration parameters only on the tuning banks." - "We never train on test scenes, and we never tune on test scenes."

Slide 18 - Checks we ran (text slide)

Say: "We verified the generator with automatic checks, all of them pass:" - Codebooks are unitary (error about $1e-15$). - Y equals FFT2 of H divided by 16 for ideal hardware (error about $1e-14$). - H has rank equal to the number of paths. - Realised SNR matches the nominal SNR (10.01 dB measured for 10 dB nominal over 3,000 scenes). - Phase error stays within the limit. Gain error stays within the limit. - Impairment label has zero mean and zero slope. - Heatmap peak is exactly 1 and stays in $[0, 1]$. - At very high SNR, the strongest bin of Y is within one cell of a true path in all tested scenes. - Angle sampler and both codebooks are identical to the authors' file `dldoa_dataset_generation.py`.

Slide 19 - Honest limitations (text slide, important - say these before someone asks)

1. **Gain error is our extension.** The two cited papers do not model gain error. The uniform-in-dB, 3 dB range is our assumption.
 2. **Impairment is redrawn for every scene and is the same for all beams.** The authors' code does the same, even though it is described as "static". Real phase shifters may depend on the beam.
 3. **Two paths can be too close to separate on the beam grid.** In L equals 3 scenes, about 3.9 percent have two paths within one beam cell (wrap-aware). About 0.35 percent of scenes have two paths that land on the *same* 32 by 32 label cell.
 4. **End-fire crowding.** 22 to 24 percent of paths fall in the single end-fire row (3.6 to 3.9 times the uniform share), caused by the cosine mapping.
 5. **Phase-ramp ambiguity.** A linear phase ramp across the array looks like an angle shift, so a small fraction of cases cannot be recovered by any method. The exact fraction depends on how you count: about 1 to 3 percent per axis at 5 degrees in our audit. That is why the label removes the ramp.
 6. **The number of paths L is given to the method** (oracle L). This follows the base paper's protocol but limits absolute claims.
 7. **Paper text and code differ.** Paper: SNR -10 to 25, L 1 to 10. Code we use: SNR -15 to 24, L 1 to 9. The $\pi/6$ separation is stated in code.
 8. **Training data is not exactly repeatable** because of OS entropy in the seed. All test banks are repeatable.
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Quick numbers cheat-sheet (memorise these)

Item	Value
Array	16 Tx antennas, 16 Rx antennas, half-wavelength ULA
Measurement	Y is 16×16 complex, $Y = W^H H F + Z$
Channel scale	$\text{sqrt}(N_t \times N_r) = 16$
Path gain	$\alpha \sim \text{CN}(0, 1/L)$, sorted strongest first
Angle range	uniform on $[0, 180 \text{ deg}]^2$, min distance $\pi/6 = 30 \text{ deg}$
Noise	each part std = $\text{sqrt}(10^{-(\text{SNR}/10)} / 2)$
Impairment	$D = g e^{j \epsilon}$; ϵ uniform $\pm \Delta$; gain uniform $\pm G \text{ dB}$ in dB
Heatmap	32×32 , Gaussian sigma = 1 cell, wrap-around, max over paths, peak = 1
Impairment label	32×2 (16 Rx + 16 Tx; phase, log-gain), mean and ramp removed
Training	L 1..9, SNR -15..24, 20% clean, $\Delta \leq 8 \text{ deg}$, $G \leq 3 \text{ dB}$, batch 256, 48,000 steps
Total training scenes	12,288,000, each used once, none stored
Test	$L = 3$, SNR -10..25 step 5, 1,000 each, 8,000 per bank
Test seeds	42 (official), 7 (dl doa), 9101 / 9102 / 9105 (T2), 9901 to 9904 (tuning)

Likely questions and short answers

Q: Why not just store a dataset? A: A simulator can make unlimited new scenes, so the network never sees the same scene twice and cannot memorise. Test sets are stored so results are reproducible.

Q: How do you know the simulator is correct? A: It reproduces the authors' generator bit-exactly for the official bank (seed 42) and a second test set (seed 7). Also the maths checks: unitary codebooks, FFT identity, correct SNR.

Q: Why a 32 by 32 heatmap and not 16 by 16? A: To locate a path between beams. Twice the resolution gives finer angle estimates.

Q: Why remove the phase ramp from the impairment label? A: A linear ramp across the array is the same as an angle shift. Data cannot tell them apart, so asking for it would be asking for the impossible.

Q: Is noise added before or after the beams? A: After the receive combiner, one independent complex Gaussian per entry of Y .

Q: Why does end-fire get so many paths? A: Angles are uniform, but beam position depends on cosine of the angle. Cosine changes slowly near 0 and 180 degrees, so many angles map to almost the same beam.

Q: Do you train on test scenes? A: No. Training seeds come from operating-system entropy. Test seeds are fixed. Tuning uses separate tuning banks.

Q: Where is the gain error from? A: It is our own extension for robustness. The papers only model phase error.

Q: What does the network get as input and what must it output? A: Input: Y as $2 \times 16 \times 16$ (real, imaginary). Outputs: the 32×32 heatmap and the 32×2 impairment vector.